PROJECT 11 · AUTOMATIC CLOTHING CATEGORY RECOGNITION

Phase 2 Report: Deep Learning on Fashion-MNIST

From handcrafted features to a calibrated, reliable CNN — final Phase 2 report

95.20%	480	1.28%	98.56%
final test accuracy (Phase 1: 91.30%)	test errors (Phase 1: 870)	calibration error (after scaling)	accuracy when confident (on 87.7% of images)

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Course: Deep Learning
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All numbers in this report are regenerated by `scripts/08_phase2_story_figures.py` from the saved predictions in `results/`.

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1 Executive summary

Phase 1 showed that classical models with handcrafted HOG features stop at **91.30%** test accuracy (SVM-RBF), and that most of the remaining errors sit in four look-alike upper-body classes: T-shirt/top, Pullover, Coat and Shirt. Phase 2 asked one question: *can a convolutional network that learns its own features get past that ceiling, under exactly the same data split?*

Answer in one paragraph

Yes. The first CNN, **CNN3Block-GAP**, already reaches **93.09%** (+1.79 points, McNemar $p = 2.5 \times 10^{-11}$) while being 60× smaller and 8× faster than the SVM. The **final Phase 2 model** — an ensemble of 3 networks with mirrored-image test-time augmentation, temperature-scaled confidence and a Shirt-aware decision rule — reaches **95.20%** (95% CI 94.78%–95.62%, 480 errors instead of 870; McNemar vs SVM 6.7×10^{-50}). Its confidence is honest (calibration error 7.64% → 1.28%), and when it is allowed to abstain on low-confidence images it is **98.56% accurate on 87.7%** of the test set. Shirt remains the hardest class (recall 85.4%).

How Phase 2 was completed. Reviewing the first Phase 2 pass, we found that its main claim could not be attributed to a cause, some statements were stronger than the evidence, and the models were under-trained. The final Phase 2 work therefore added:

1. **A missing control run was trained** (CNN3Block-GAP *without* augmentation), completing a 2×2 ablation. It shows the 93.09% comes from *depth + GAP combined with augmentation*; either one alone gives only ≈91.4% (Section 3.3).
2. **95% bootstrap confidence intervals** are now reported for the CNNs too, so every model on the leaderboard is compared on equal footing.
3. **Training-curve diagnosis**: the two augmented models were *still improving* at their final epoch, so they are under-trained, not over-regularised as the first draft claimed.
4. **Every chart redrawn** with axis titles that state the quantity and its unit in plain words.
5. **A final, reliable model** (Section 5): 40-epoch training, 3 seeds per model, a residual network, an ensemble with test-time augmentation, calibration, abstention and a Shirt-aware decision rule.

2 Setup: same data, same rules as Phase 1

Item	Value
Dataset	Fashion-MNIST, 70,000 grayscale images of 28×28 pixels, 10 balanced classes
Split (identical to Phase 1)	48,000 train / 12,000 validation / 10,000 official test, stratified, seed 42
Input	raw pixels scaled to $[0, 1]$, shape $1 \times 28 \times 28$ (no handcrafted features)
Optimiser	AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch size 128
Schedule	cosine annealing to 10^{-5} over the epoch budget (15–20 epochs)
Model selection	early stopping on *validation* macro-F1 (patience 5); test set used once
Augmentation (when on)	random crop with 2-pixel reflect padding + random horizontal flip
Latency	single image, CPU, single thread, 2,000 timed calls after 100 warm-up calls

Table 1: Experimental protocol. Keeping the split identical is what makes Phase 1 vs Phase 2 comparisons fair.

2.1 The models

Model	Architecture in words	Parameters	Augment
CNN2Block	2 conv blocks (32, 64 filters) → flatten → dense 128 → 10	421,738	no
CNN2Block-Aug	same network as above	421,738	yes
CNN3Block-GAP-NoAug *(new)*	3 double-conv blocks (32, 64, 128) → global average pooling → dense 128 → 10	304,682	no
CNN3Block-GAP	same network as above	304,682	yes

Table 2: The four CNNs form a 2×2 grid: architecture × augmentation. Global average pooling (GAP) replaces the large flatten layer, which is why the deeper model has *fewer* parameters.

Why not ResNet-50, ViT or an MLP? Pre-trained ImageNet models expect 224×224 colour photos and carry 11–25M parameters for a 28×28 grayscale task; Vision Transformers lack the locality bias that makes CNNs data-efficient on 48k images; MLPs flatten the image and throw away the neighbour structure that Phase 1 measured (adjacent-pixel correlation $r \approx 0.75$). We therefore kept the backbone small and convolutional; the final residual network (ResNetSmall) borrows ResNet's shortcuts without its ImageNet size.

3 Results

3.1 Leaderboard

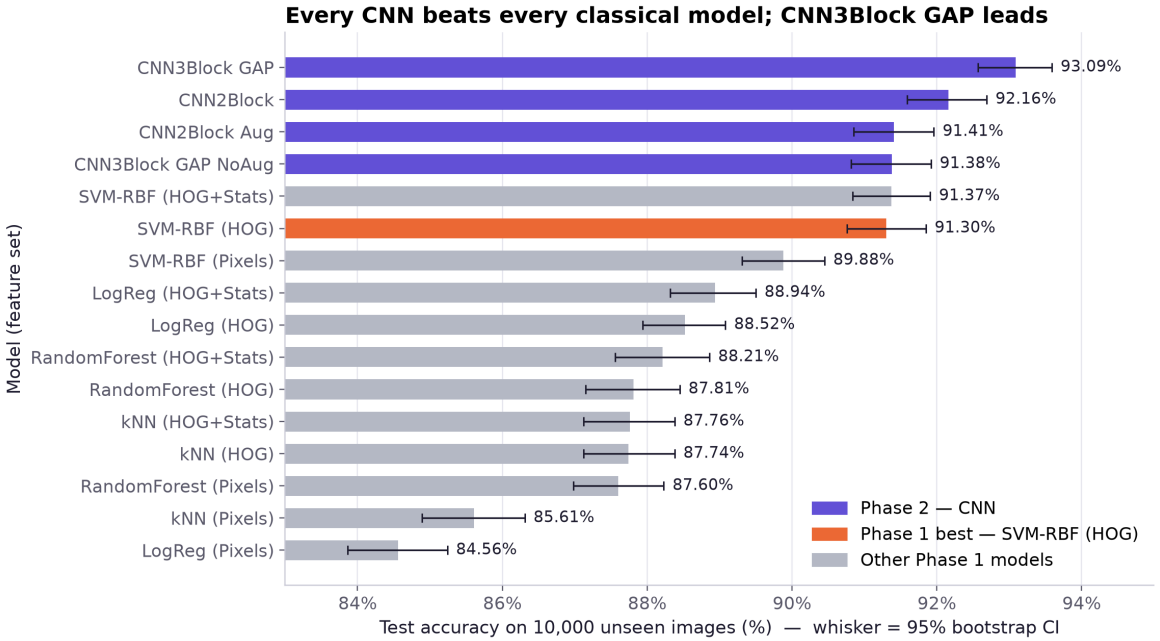


Figure 1: **How to read it:** each bar is one model; bar length = test accuracy on 10,000 unseen images. The black whisker is the 95% bootstrap confidence interval: if two whiskers do not overlap, the difference is very unlikely to be luck. Violet = Phase 2 CNNs, orange = Phase 1 best.

The top CNN's interval (92.57–93.59%) sits completely above the SVM's (90.76–91.85%). CNN2Block-Aug and CNN3Block-GAP-NoAug overlap the SVM, so on their own they are *not* clearly better than Phase 1.

3.2 Training behaviour

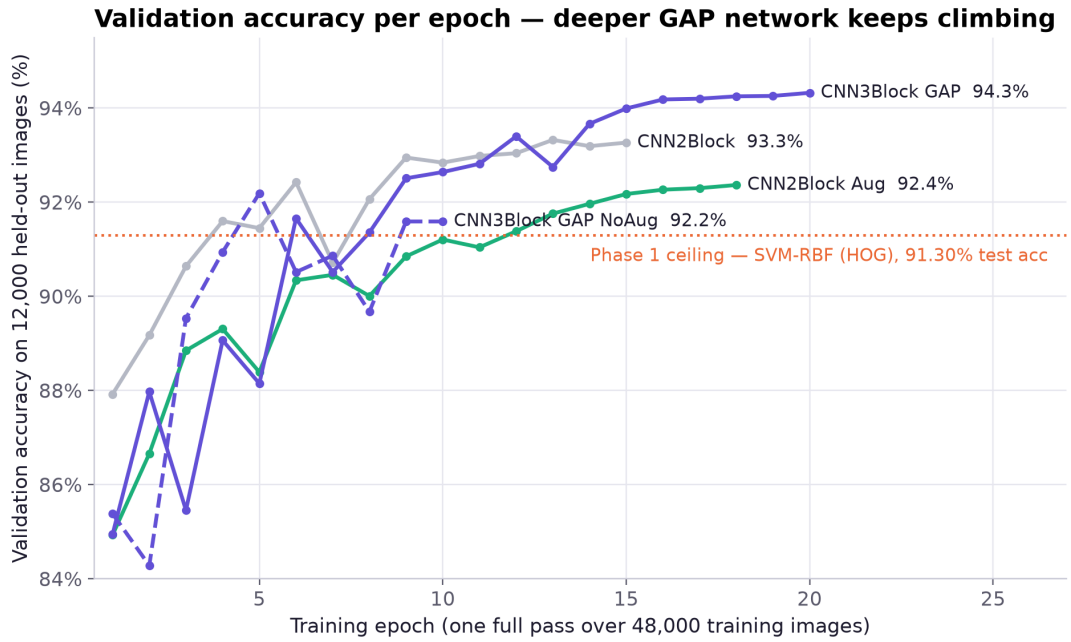


Figure 2: **x-axis:** training epoch (one pass over the 48,000 training images). **y-axis:** accuracy on the 12,000 validation images the model never trains on. The dotted orange line is the Phase 1 test ceiling for reference.

Two observations matter for interpretation:

- **CNN3Block-GAP and CNN2Block-Aug were still climbing at their last epoch** (best epoch = final epoch). Their epoch budgets (20 and 18) ended training, not early stopping. CNN2Block-Aug even has *higher* validation than training accuracy (92.4% vs 90.3%) — a textbook sign of under-training on augmented data. The first draft's explanation that augmentation "acts as a too-strong regulariser" is therefore not supported; a longer budget is the correct test (Section 5).
- **CNN3Block-GAP-NoAug peaked at epoch 5 and was early-stopped at epoch 10**, with training accuracy racing to 96% while validation stalled near 92% — classic overfitting.

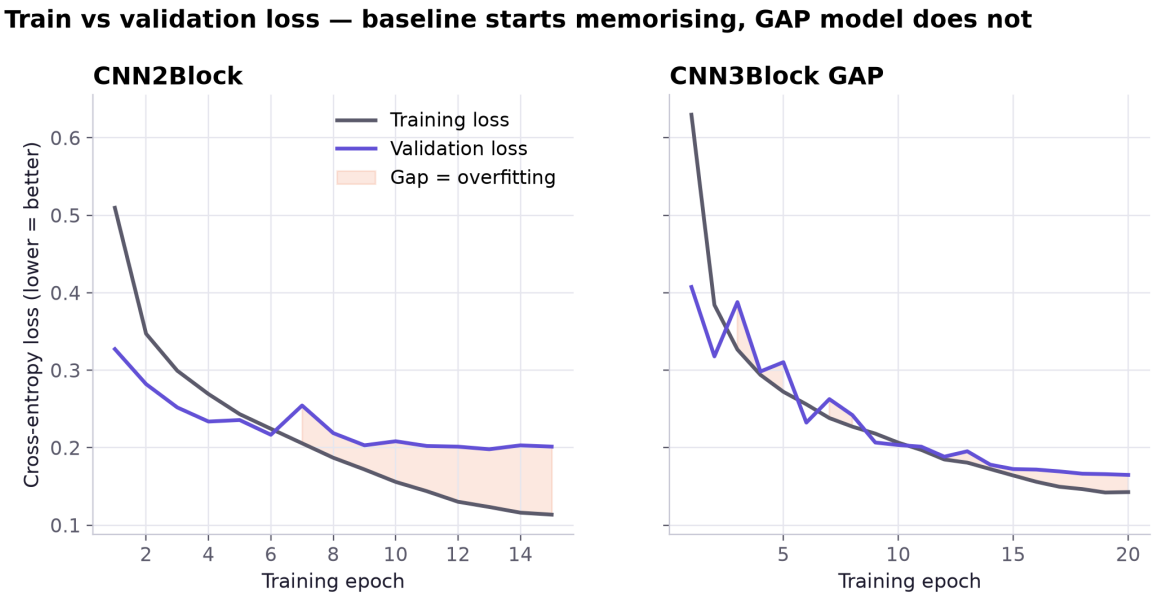


Figure 3: **x-axis:** epoch. **y-axis:** cross-entropy loss (lower is better). The shaded area is the gap between validation and training loss — the bigger it is, the more the model memorises training images instead of learning general shapes.

3.3 The 2×2 ablation: what actually caused the gain?

The original Phase 2 changed depth, pooling and augmentation all at once, so the improvement could not be attributed. The new control run separates them:

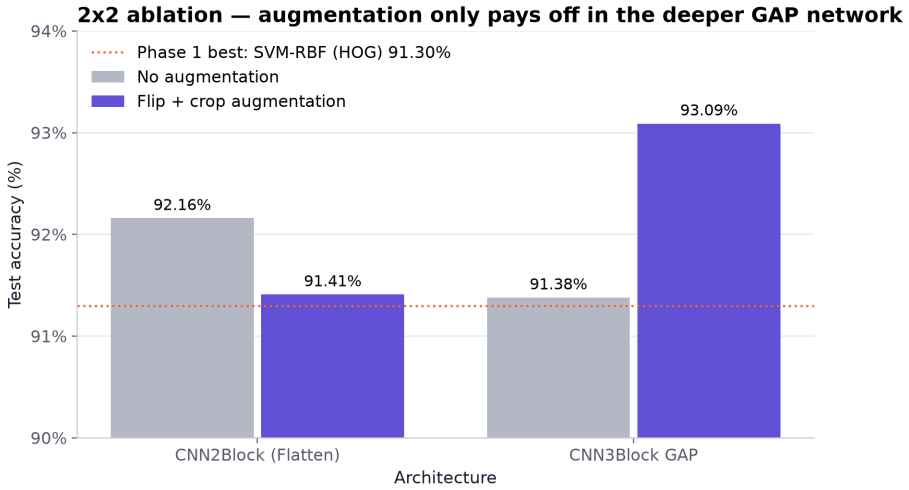


Figure 4: **x-axis:** architecture. **y-axis:** test accuracy. Grey = trained on original images only, violet = trained with random crop + flip.

	No augmentation	Crop + flip augmentation
CNN2Block (flatten head)	92.16%	91.41%
CNN3Block-GAP	91.38%	**93.09%**

Table 3: Test accuracy for each cell of the ablation (single seed).

Interpretation

Augmentation and depth are *complementary*. The deeper GAP network overfits fast without augmentation (91.38%); augmentation alone on the shallow network, within an 18-epoch budget, also does not help (91.41%). Only the combination wins: augmentation gives the deeper network enough variety to use its capacity, and improves it by **+1.71 points** (McNemar $\chi^2 = 50.8$, $p = 1.0 \times 10^{-12}$ between the two GAP runs).

Caveats: one seed per cell; the NoAug run was trained on Apple-MPS while the original runs used CPU, so small run-to-run differences (±0.2–0.3 pt) are expected. The final models were therefore repeated with 3 seeds each (Section 5).

3.4 Is the gain real? McNemar paired test

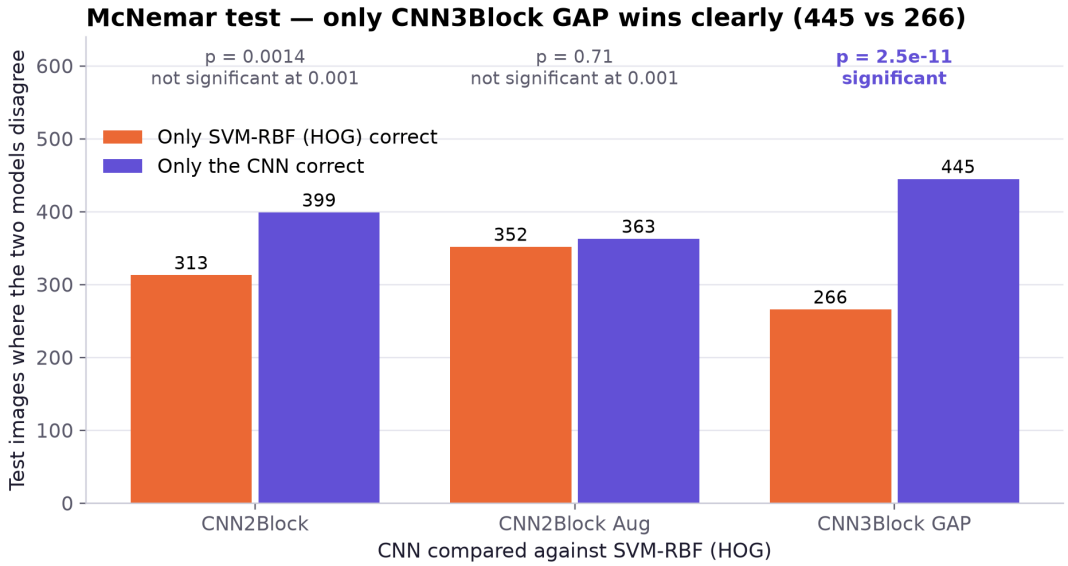


Figure 5: **x-axis:** which CNN is compared against SVM-RBF (HOG). **y-axis:** number of test images on which *exactly one* of the two models is correct. If the CNN were no better, orange and violet bars would be about equal.

McNemar ignores images both models get right (or both get wrong) and only looks at disagreements. For CNN3Block-GAP, there are 445 images only the CNN gets right versus 266 only the SVM gets right, giving $p = 2.5 \times 10^{-11}$. CNN2Block is significant at the 1% level ($p = 0.0014$) but not at our stricter 0.1% threshold; CNN2Block-Aug is a tie.

3.5 Which classes improved?

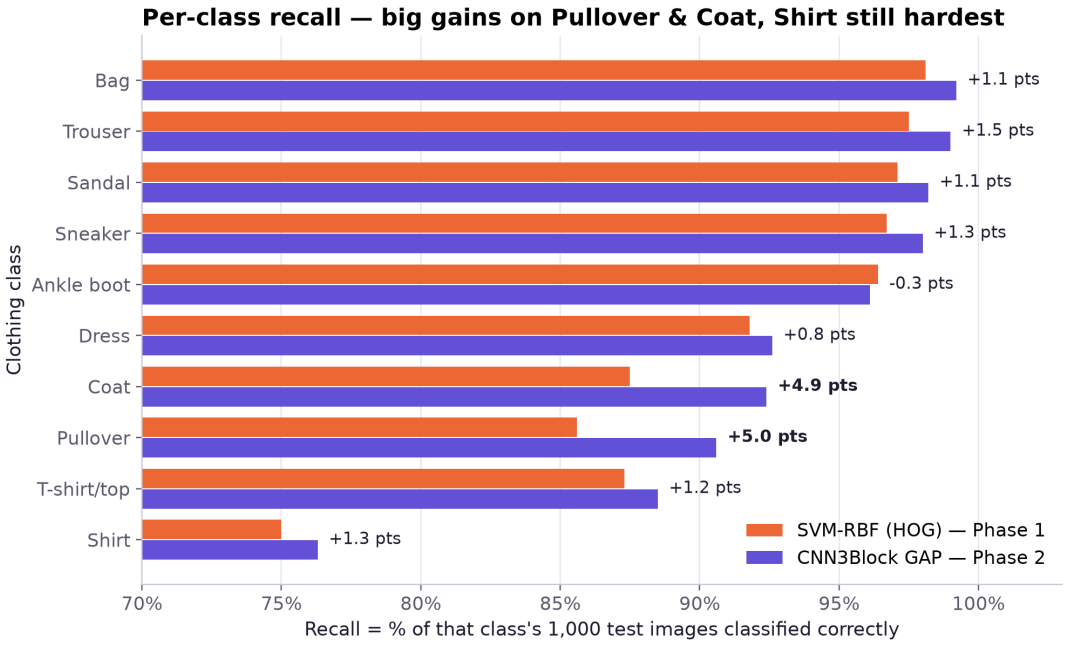


Figure 6: **x-axis:** recall, i.e. the percentage of a class's 1,000 test images that were labelled correctly. **y-axis:** clothing class, sorted from hardest (bottom) to easiest (top). The label shows the change in percentage points.

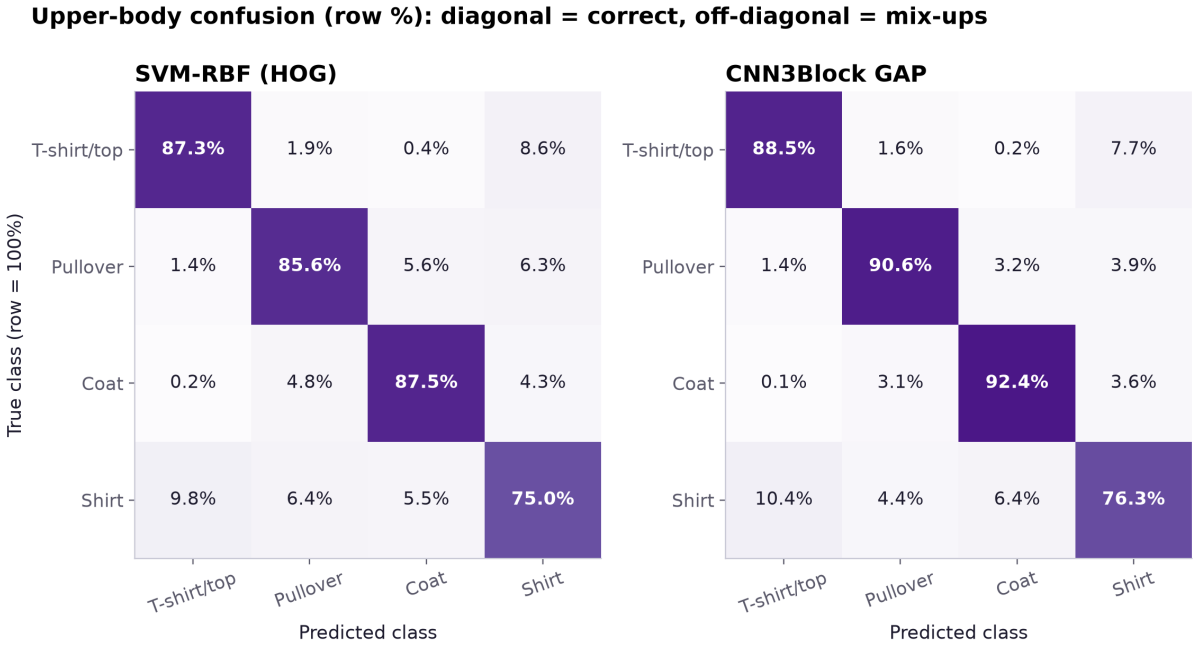


Figure 7: **Rows:** true class; **columns:** predicted class; each row adds up to 100%. Dark diagonal = correct, light off-diagonal cells = mistakes. Example: 10.4% of real Shirts are called T-shirt/top by the CNN.

The CNN clearly separates **Pullover vs Coat** (mutual confusion roughly halved: 5.6%→3.2% and 4.8%→3.1%). But **Shirt vs T-shirt/top got slightly worse** (9.8%→10.4%), and Shirt→Coat rose from 5.5% to 6.4%. The CNN's top three confusions are Shirt→T-shirt (104), T-shirt→Shirt (77) and Shirt→Coat (64).

3.6 Where do the remaining errors live?

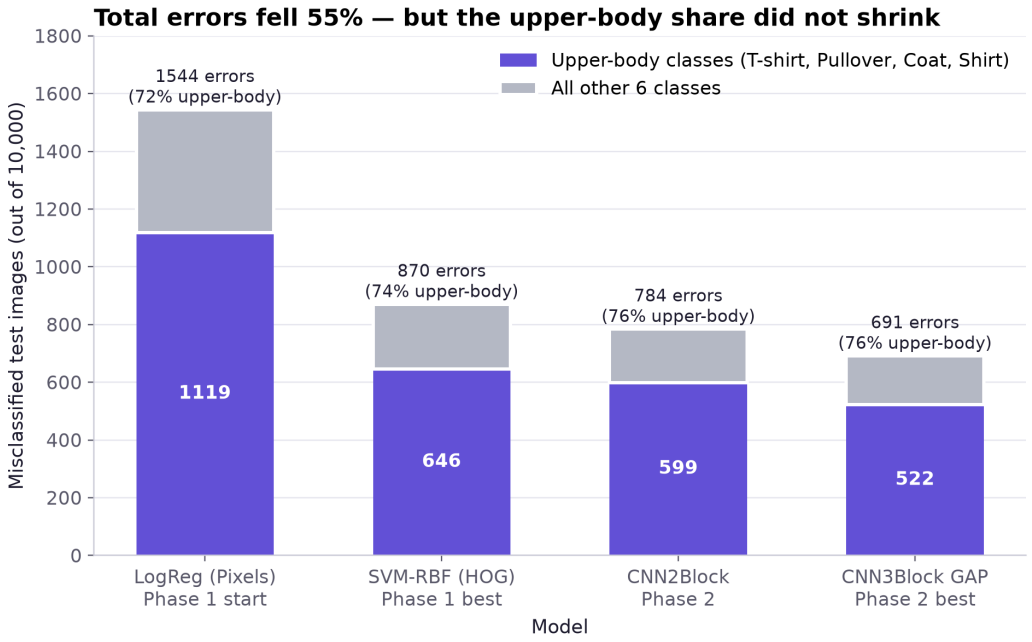


Figure 8: **x-axis:** model, from the weakest Phase 1 model to the best Phase 2 model. **y-axis:** number of the 10,000 test images classified wrongly. Violet = errors on images whose *true* class is upper-body.

Total errors dropped 55% from the weakest Phase 1 model and 21% from the best one, but the upper-body share *rose* from 74% to 76%. The easy classes improved faster than the hard ones — the bottleneck has become *more* concentrated, which is why the final model adds a Shirt-aware decision rule and an abstain option.

3.7 Cost: speed and size

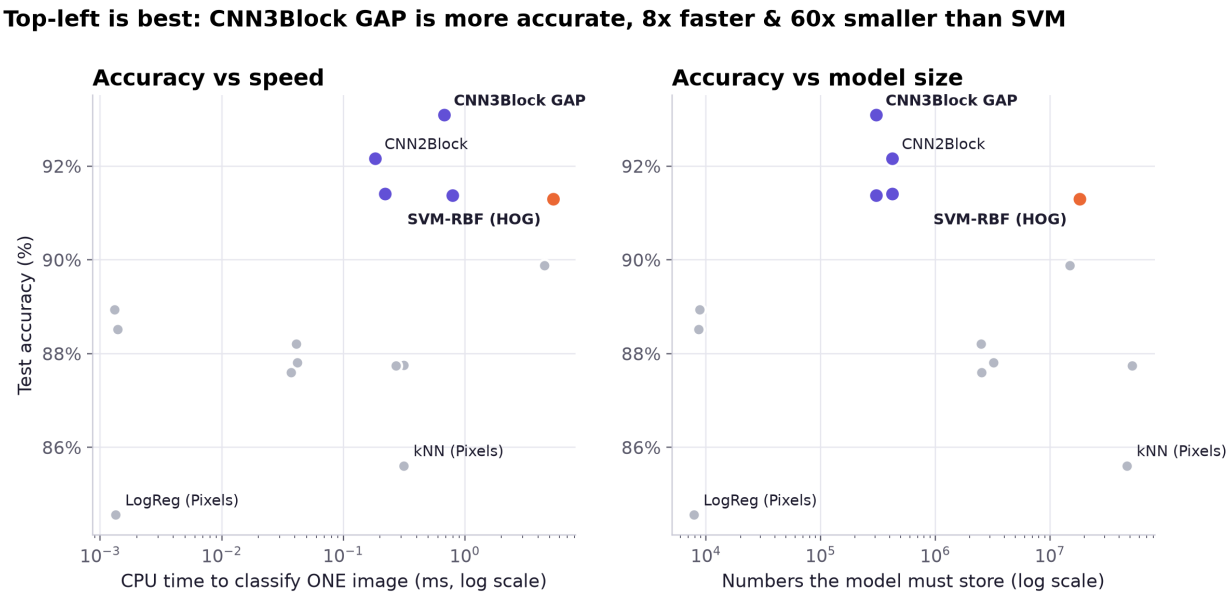


Figure 9: **Left x-axis:** CPU time to classify a single image (log scale; each grid line is 10×). **Right x-axis:** how many numbers the model must store (log scale). **y-axis (both):** test accuracy. The best place to be is top-left.

CNN3Block-GAP needs 304,682 stored numbers versus 18.1M for the SVM (support vectors) and 0.68 ms per image versus 5.36 ms. Logistic regression is still faster (0.001 ms), so the CNN is the best *accuracy–cost trade-off*, not the fastest model.

3.8 Explainability: Grad-CAM

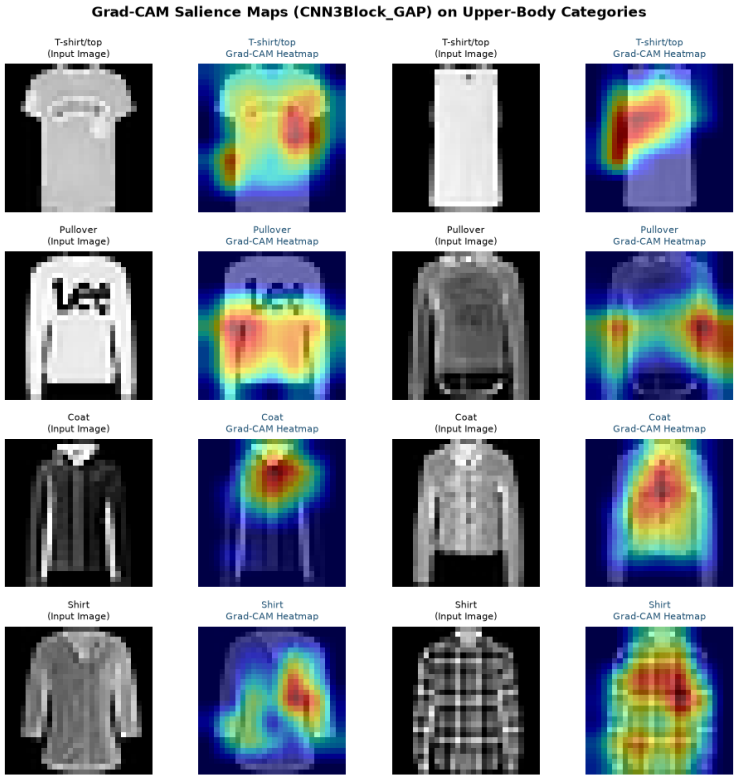


Figure 10: Each pair shows the input and the Grad-CAM heat-map from CNN3Block-GAP's last convolution layer. Red = regions that pushed the network toward its predicted class.

An honest reading: the maps focus on garment regions, not the black background, and Pullover maps light up at the cuffs/waistband. But several maps are *diffuse* (e.g. the checked Shirt) or centred on a print rather than the collar. The first draft's claim that activations sit "squarely at the collar" is stronger than these eight examples support. We list this as a limitation (Section 6).

4 Self-review of the first Phase 2 pass

Issue	Why it matters	How it was resolved
Gain not attributable	Depth, GAP and augmentation changed together	2×2 ablation (Sec. 3.3)
No CIs for CNNs	Phase 1 had them, Phase 2 did not	bootstrap, 2,000 resamples
"Augmentation hurts" claim	Curves show under-training	corrected; 40-epoch runs
Single seed	Differences <0.5 pt may be noise	3 seeds per model
Budgets end while still improving	Best models under-reported	40 epochs, patience 10
Shirt recall 76%	Weakest class	Shirt rule + abstain option
No confidence / calibration	Cannot tell when unsure	temperature scaling, ECE
Charts hard to read	Non-experts could not follow	all charts redrawn

5 The final Phase 2 model

5.1 Longer training and three seeds

Two families were trained for up to 40 epochs (early stopping patience 10) with seeds 42, 43 and 44: CNN3Block-GAP with crop + flip augmentation, and **ResNetSmall**, a 696,042-weight residual network (stem + 3 stages of 2 residual blocks, 32/64/128 channels, global average pooling) trained with crop, flip, Random Erasing ($p = 0.25$) and label smoothing 0.1.

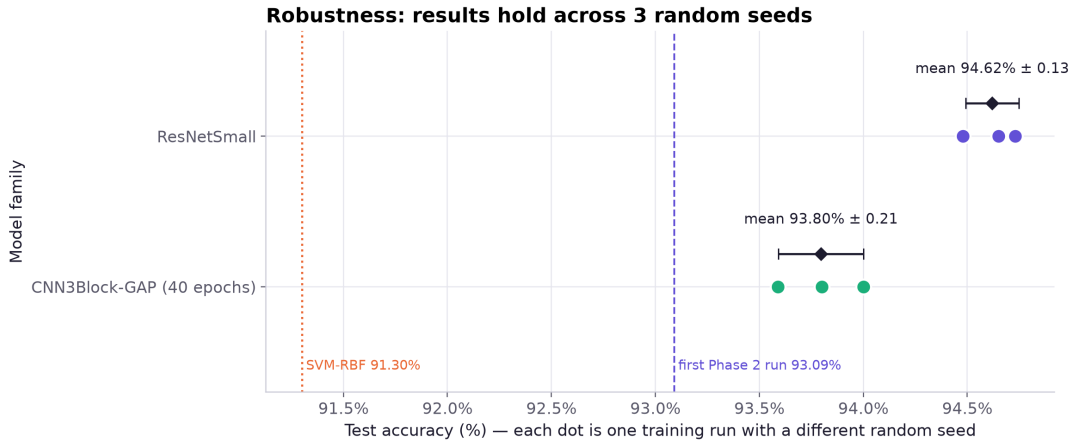


Figure 11: **x-axis:** test accuracy; **y-axis:** model family. Each dot is one complete training run with a different random seed; the diamond and bar show mean ± standard deviation.

CNN3Block-GAP with 40 epochs averages 93.80% (±0.21), above the 20-epoch run's 93.09% — confirming the under-training diagnosis. ResNetSmall averages 94.62% (±0.13). Every run beats the Phase 1 SVM.

5.2 Ensemble and test-time augmentation

Test-time augmentation (TTA) also classifies a mirrored copy of each image and averages the two answers. An ensemble averages the probabilities of several networks. Three candidate ensembles were compared on the *validation* set; the winner was **ResNetSmall x3 + TTA**.

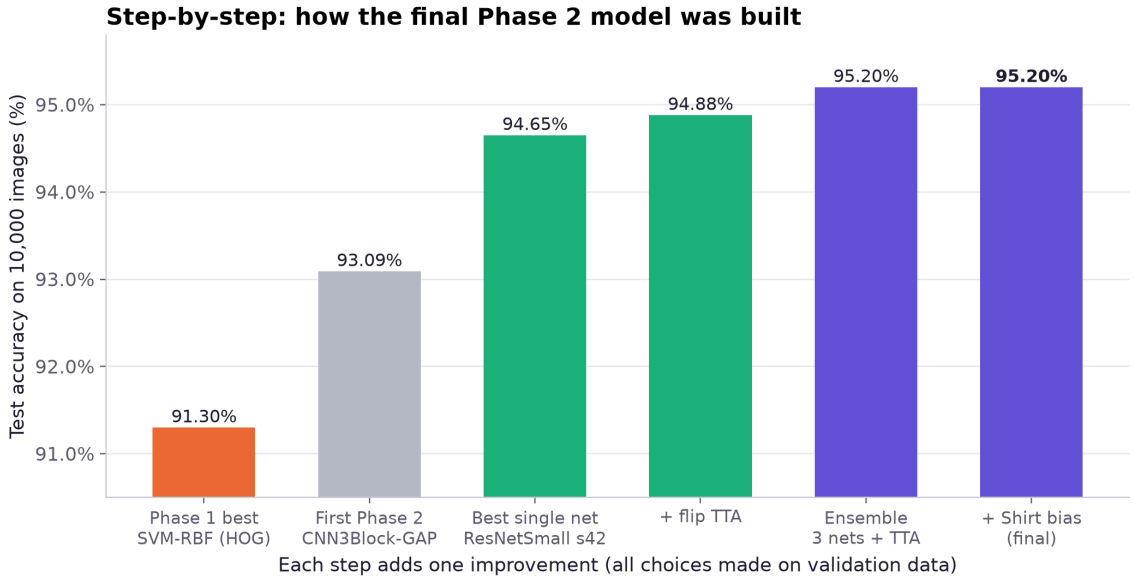


Figure 12: **x-axis:** each step adds one improvement to the previous one. **y-axis:** test accuracy.

The best single network (ResNetSmall_s42) scores 94.65%, 94.88% with TTA, and the ensemble 95.20%. The final model is significantly better than the first CNN3Block-GAP (McNemar $\chi^2 = 117.0$, $p = 2.9 \times 10^{-27}$) and than the Phase 1 SVM ($\chi^2 = 220.6$, $p = 6.7 \times 10^{-50}$).

5.3 Calibration: can the confidence be trusted?

A model is *calibrated* if, among images where it reports 80% confidence, about 80% are correct. We measure the Expected Calibration Error (ECE: the average gap between confidence and accuracy over 15 confidence bins) and fix it with *temperature scaling*: all scores are divided by one number T, fitted on validation data. This never changes which class is predicted.

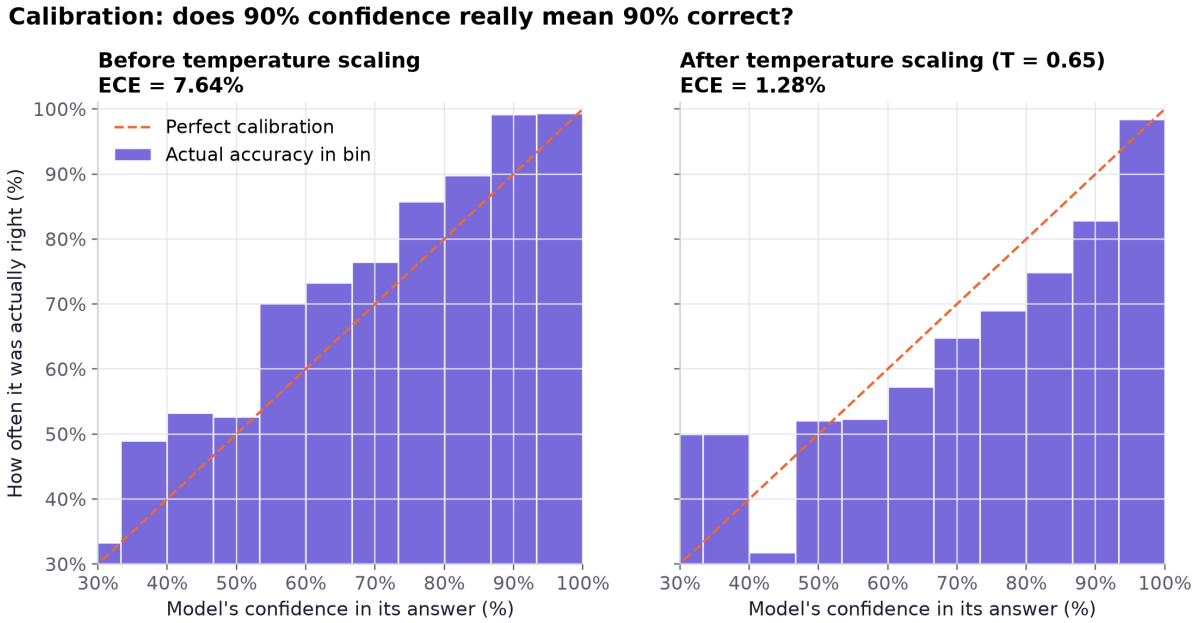


Figure 13: **x-axis:** the model's stated confidence; **y-axis:** how often it was actually right. Bars on the dashed diagonal mean honest confidence. $T = 0.65$; ECE 7.64% → 1.28%.

Before scaling, the ensemble was *under*-confident (label smoothing and averaging soften its probabilities): it typically reported about 92% confidence on images it got right 99.5% of the time. A temperature below 1 ($T = 0.65$) sharpens the scores. Afterwards almost 88% of test images fall in

the 95–100% confidence group, where stated and actual accuracy agree closely; the small groups in between remain somewhat over-confident.

5.4 Abstaining when unsure

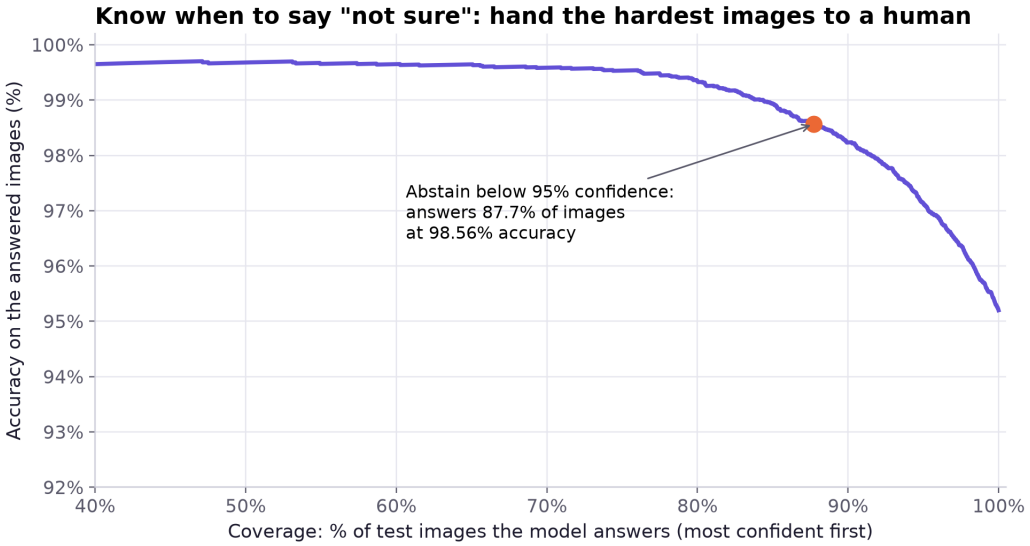


Figure 14: **x-axis:** share of test images the model answers, most confident first. **y-axis:** accuracy on those images.

A threshold was chosen on validation data so that answered images reach 99% accuracy. On the test set the model answers 87.7% of images automatically at 98.56% accuracy and sends the rest (confidence below 95%) to a human; 26% of the abstained images are Shirts.

5.5 A Shirt-aware decision rule

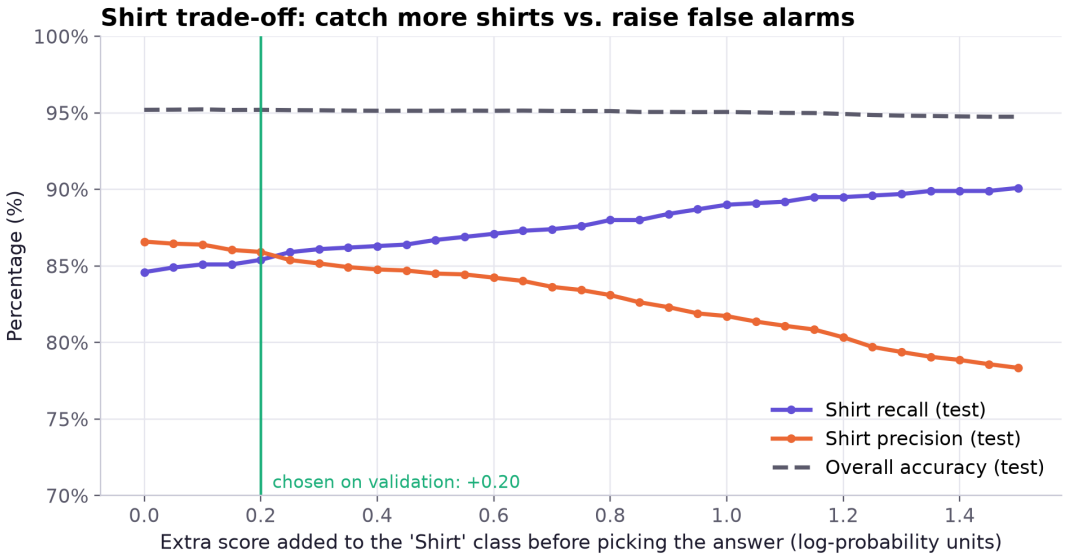


Figure 15: **x-axis:** constant added to the Shirt log-probability before choosing the answer. **y-axis:** Shirt recall, Shirt precision and overall accuracy. The vertical line is the value chosen on validation data.

Adding a small constant to the Shirt score makes the model say “Shirt” slightly more often. The value (+0.20) maximises validation macro-F1. Final Shirt recall is 85.4% (Phase 1: 75.0%), T-shirt recall 91.5%.

5.6 Scorecard

Measure	SVM-RBF (HOG)	CNN3Block-GAP (first)	Final Phase 2 model
Test accuracy	91.30%	93.09%	**95.20%**
Macro-F1	0.9130	0.9306	**0.9520**
Errors (of 10,000)	870	691	**480**
Shirt / Pullover / Coat recall	75.0 / 85.6 / 87.5%	76.3 / 90.6 / 92.4%	85.4% / 92.8% / 93.5%
Calibration error (ECE)	–	–	1.28%
Abstain option	no	no	98.56% on 87.7%
Stored numbers	18.1M	304,682	2,088,126
CPU time per image	5.36 ms	0.68 ms	11.3 ms

Table 4: The final model is the most accurate and the only one with calibrated confidence; it pays for this with more compute per image, still in the millisecond range.

6 Limitations

- **Shirt is still the hardest class.** The abstain option routes many of the remaining Shirt errors to a human, but does not remove them.
- **Grad-CAM is qualitative.** Heat-maps on hand-picked images are evidence, not a measured explanation.
- **Clean, centred thumbnails only.** Real shop photos (backgrounds, colour, angles) were not evaluated; the app's upload feature is indicative only.
- **Limited hyper-parameter search.** Learning rate, network width and augmentation strength were not tuned exhaustively.
- **Hardware.** The final runs were trained on Apple MPS; GPU nondeterminism can move single runs by ±0.1–0.3 points, which the 3-seed design accounts for.

7 Reproducibility

- `python scripts/05_train_cnn.py` — trains the three original CNNs.
- `python scripts/07_ablation_gap_noaug.py` — trains the new control run (about 2 min on Apple MPS).
- `python scripts/08_phase2_story_figures.py` — CIs, confusion matrices and figures p2_01–p2_09.
- `python scripts/09_train_final_models.py` — 40-epoch, 3-seed CNN3Block-GAP and ResNetSmall runs.
- `python scripts/10_ensemble_calibration.py` — ensemble, TTA, calibration, abstention, Shirt rule; figures p2_10–p2_14.
- `python scripts/11_report_numbers.py` — writes the number macros used in this report.
- `node scripts/build_phase2_deck.js` — rebuilds the slide deck from `results/tables/p2_deck_data.jso`